

Overview in General

This file contains a general overview of the data in the graph including node labels and relationships types.

References

- [jqassistant](#)
- [Neo4j Python Driver](#)

Node Labels

Table 1a - Highest node count by label combination

Lists the 30 label combinations with the highest number of nodes. The labels with the lowest node count are listed in table 1b. The total list would sum up to the total number of labels (100%).

The whole table can be found in the CSV report `Node_label_combination_count`.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Git, Update, Change]	58793	52.230731
1	[Git, Change, Create]	17294	15.363704
2	[Git, Commit]	11298	10.036957
3	[Git, Change, Delete]	8782	7.801784
4	[File, Git]	5905	5.245905
5	[Git, Change, Rename]	3355	2.980527
6	[Git, Tag]	1758	1.561778
7	[Author, Git, Person]	1267	1.125582
8	[Json, Key]	668	0.593440
9	[Json, Value, Scalar]	603	0.535695
10	[Committer, Git, Person]	370	0.328702
11	[NPM, Dependency]	338	0.300274
12	[Type, TS, Primitive]	293	0.260296
13	[Type, TS, Declared]	277	0.246082
14	[TS, ExternalDeclaration]	215	0.191002
15	[Git, Change, Copy]	138	0.122597
16	[Type, TS, Literal]	136	0.120820
17	[Json, Value, Object]	133	0.118155
18	[Type, TS, Union]	120	0.106606
19	[Type, TS, ObjectMember]	103	0.091504
20	[NPM, Script]	91	0.080843
21	[TS, Property]	65	0.057745
22	[TS, Function]	47	0.041754
23	[Git, Branch]	46	0.040866
24	[Type, TS, FunctionParameter]	40	0.035535
25	[Type, Object, TS]	39	0.034647
26	[File, Directory]	34	0.030205
27	[Type, TS, Function]	34	0.030205
28	[TS, Parameter]	33	0.029317
29	[Package, File, Json, NPM]	29	0.025763

Chart 1a - Highest node count by label combination

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in Chart 1b.

<Figure size 640x480 with 0 Axes>

Nodes per label combination (more than 0.5% overall)

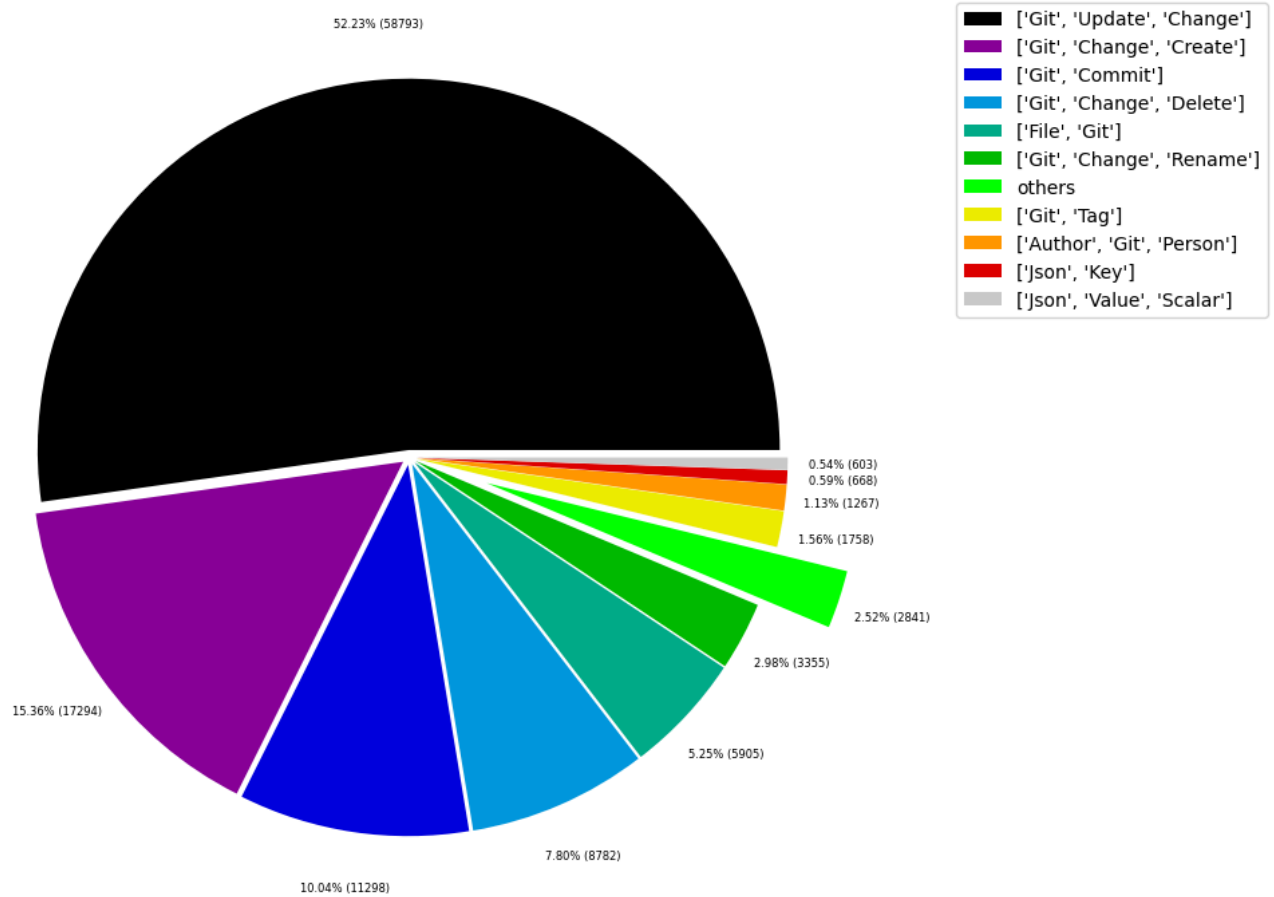


Table 1b - Lowest node count by label combination

Lists the 30 label combinations with the lowest number of nodes until they reach 0.5% of the total node count, which are shown above.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Analyze, Task, JQAssistant]	1	0.000888
1	[NPM, PublishConfig]	1	0.000888
2	[File, TS, Scan]	1	0.000888
3	[TS, Method]	1	0.000888
4	[Repository, File, Git]	1	0.000888
5	[TS, Constructor]	1	0.000888
6	[Value, TS, ObjectMember]	1	0.000888
7	[TS, Class]	1	0.000888
8	[TS, Enum]	2	0.001777
9	[Value, Object, TS]	3	0.002665
10	[Type, TS, Tuple]	3	0.002665
11	[Value, TS, Function]	4	0.003554
12	[TS, TypeParameter]	4	0.003554
13	[Value, TS, Complex]	5	0.004442
14	[Repository, NPM]	5	0.004442
15	[NPM, Engine]	6	0.005330
16	[Project, TS]	6	0.005330
17	[File, Local]	6	0.005330
18	[Type, TS, TypeParameterReference]	6	0.005330
19	[Value, TS, Member]	6	0.005330
20	[File, TS, Local, Module]	6	0.005330
21	[Value, TS, Call]	6	0.005330
22	[TS, EnumMember]	8	0.007107
23	[Type, TS, NotIdentified]	11	0.009772
24	[TS, ExternalModule]	11	0.009772
25	[Json, Value, Array]	12	0.010661
26	[Value, TS, Declared]	13	0.011549
27	[TS, TypeAlias]	16	0.014214
28	[File, Directory, Local]	16	0.014214
29	[TS, Interface]	17	0.015103

Chart 1b - Lowest node count by label combination

Shows the lowest (less than 0.5% overall) node count label combinations. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Nodes per label combination (less than 0.5% overall)

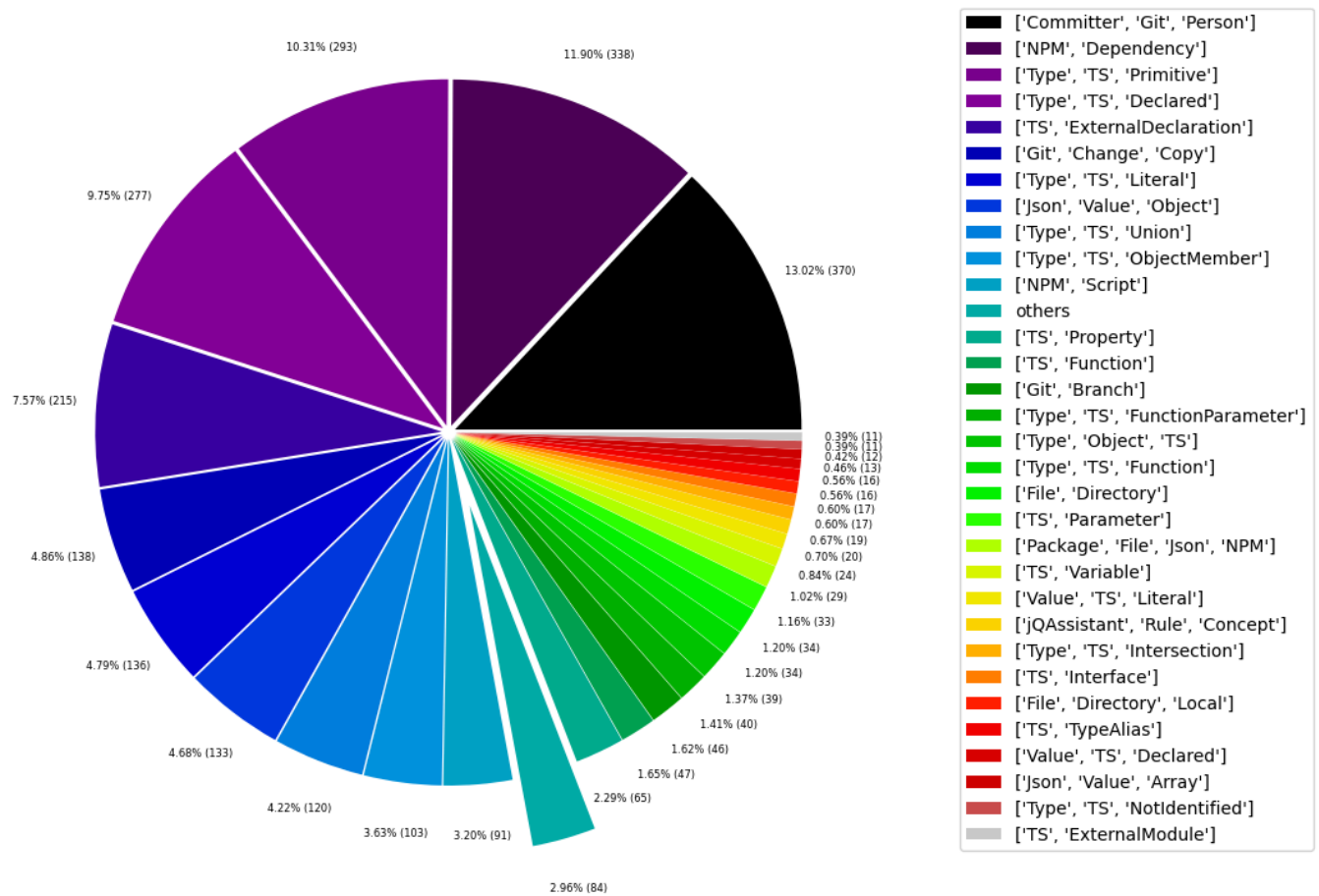


Table 1c - Highest node count by single label

Lists the 40 labels with the highest number of nodes. Doesn't sum up to the total number of nodes or 100% because one node can have multiple labels. Helps to identify commonly used labels.

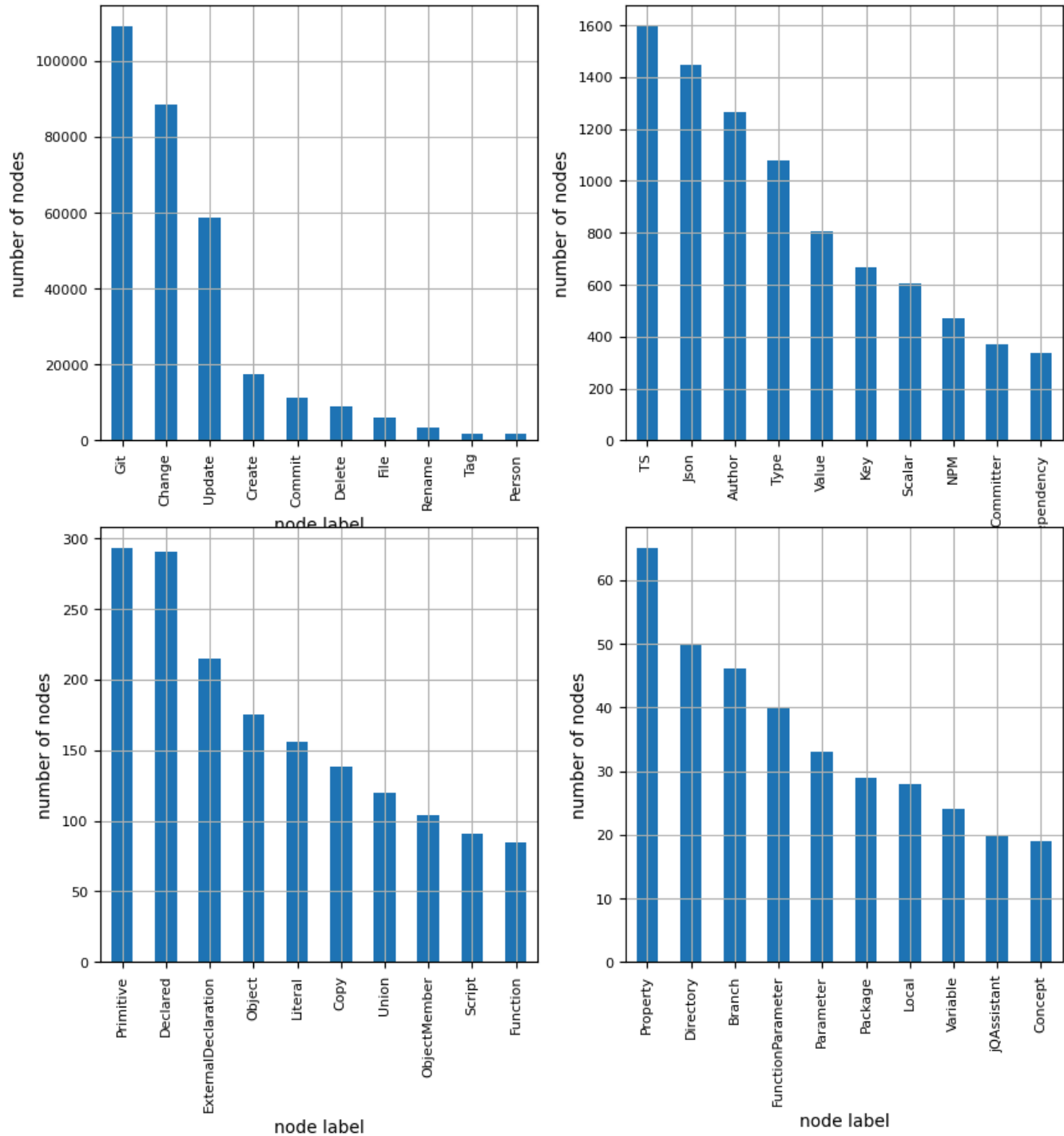
	nodeLabel	nodesWithThatLabel	nodesWithThatLabelPercent
0	Git	109007	96.840020
1	Change	88362	78.499343
2	Update	58793	52.230731
3	Create	17294	15.363704
4	Commit	11298	10.036957
5	Delete	8782	7.801784
6	File	5998	5.328524
7	Rename	3355	2.980527
8	Tag	1758	1.561778
9	Person	1637	1.454284
10	TS	1595	1.416972
11	Json	1445	1.283714
12	Author	1267	1.125582
13	Type	1079	0.958566
14	Value	806	0.716037
15	Key	668	0.593440
16	Scalar	603	0.535695
17	NPM	470	0.417540
18	Committer	370	0.328702
19	Dependency	338	0.300274
20	Primitive	293	0.260296
21	Declared	290	0.257631
22	ExternalDeclaration	215	0.191002
23	Object	175	0.155467
24	Literal	156	0.138588
25	Copy	138	0.122597
26	Union	120	0.106606
27	ObjectMember	104	0.092392
28	Script	91	0.080843
29	Function	85	0.075513
30	Property	65	0.057745
31	Directory	50	0.044419
32	Branch	46	0.040866
33	FunctionParameter	40	0.035535
34	Parameter	33	0.029317
35	Package	29	0.025763
36	Local	28	0.024875
37	Variable	24	0.021321
38	jqAssistant	20	0.017768
39	Concept	19	0.016879

Chart 1c - Highest node count by label

Shows the 40 labels with the highest number of nodes.

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Node count by label



Relationship Types

Table 2a - Highest relationship count by type

Lists the 30 relationship types with the highest number of occurrences. The whole table can be found in the CSV report `Relationship_type_count`.

Total number of relationships: 336124

	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	CONTAINS_CHANGE	88362	26.288513
1	MODIFIES	88362	26.288513
2	UPDATES	58793	17.491461
3	COMMITTED	22596	6.722519
4	CREATES	20787	6.184325
5	HAS_PARENT	12383	3.684057
6	DELETES	12137	3.610870
7	HAS_COMMIT	11298	3.361260
8	HAS_FILE	5905	1.756792
9	RENAMES	3355	0.998144
10	HAS_NEW_NAME	1771	0.526889
11	HAS_TAG	1758	0.523021
12	ON_COMMIT	1758	0.523021
13	HAS_AUTHOR	1267	0.376944
14	DEPENDS_ON	887	0.263891
15	HAS_KEY	668	0.198736
16	HAS_VALUE	668	0.198736
17	CONTAINS	596	0.177316
18	HAS_COMMITTER	370	0.110078
19	OF_TYPE	339	0.100856
20	EXPORTS	309	0.091930
21	REFERENCES	197	0.058609
22	DECLARES	186	0.055337
23	DECLARES_DEV_DEPENDENCY	169	0.050279
24	DECLARES_DEPENDENCY	161	0.047899
25	COPIES	138	0.041056
26	HAS_MEMBER	104	0.030941
27	HAS_TYPE_ARGUMENT	94	0.027966
28	DECLARES_SCRIPT	91	0.027073
29	RETURNS	82	0.024396

Chart 2a - Highest relationship count by type

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in the second chart.

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Relationship types (more than 0.5% overall)

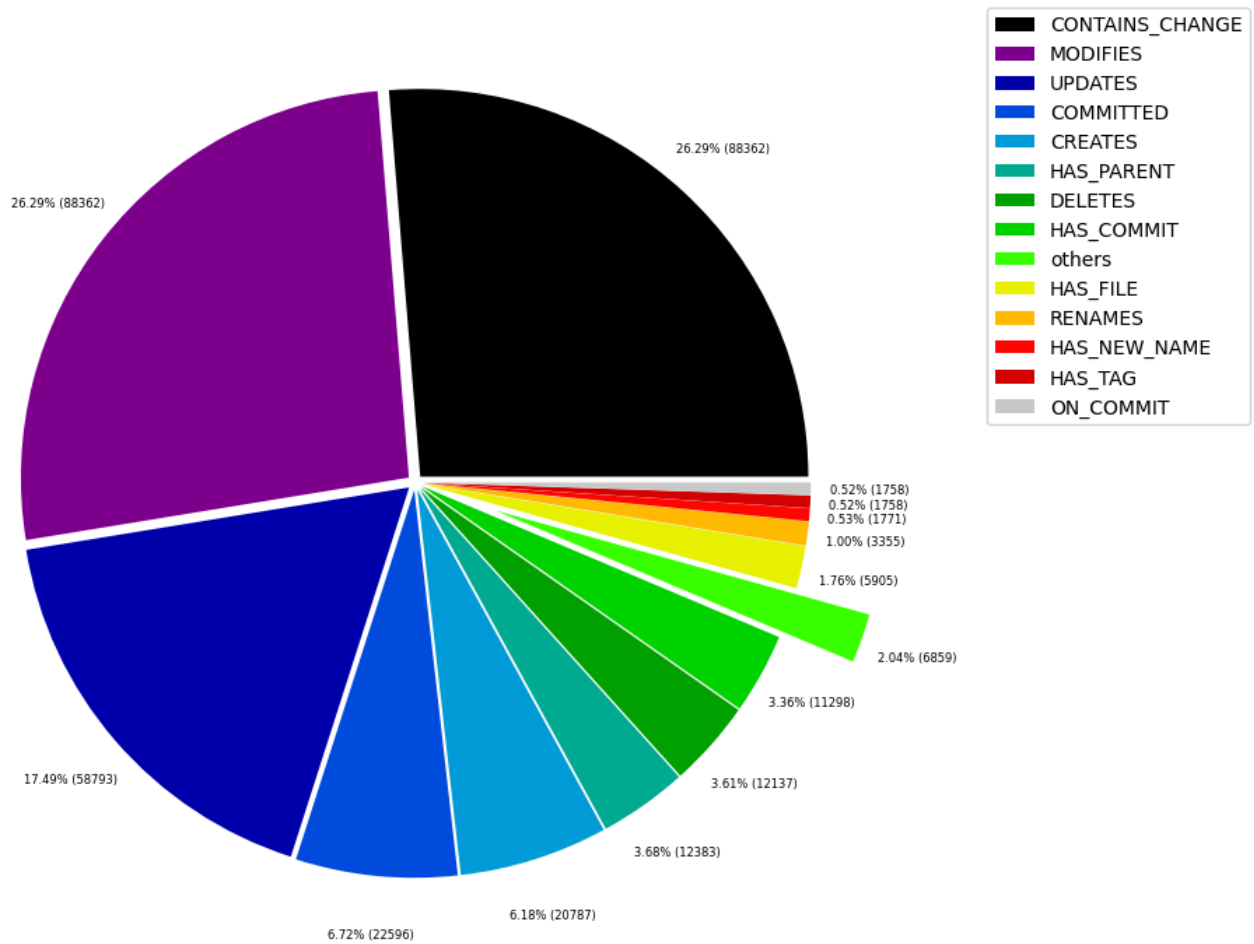


Table 2b - Lowest relationship count by type

Lists the 30 relationships type with the lowest number of occurrences up to 0.5% of the total node count. This is essentially breaking down the "others" slice from the chart above.

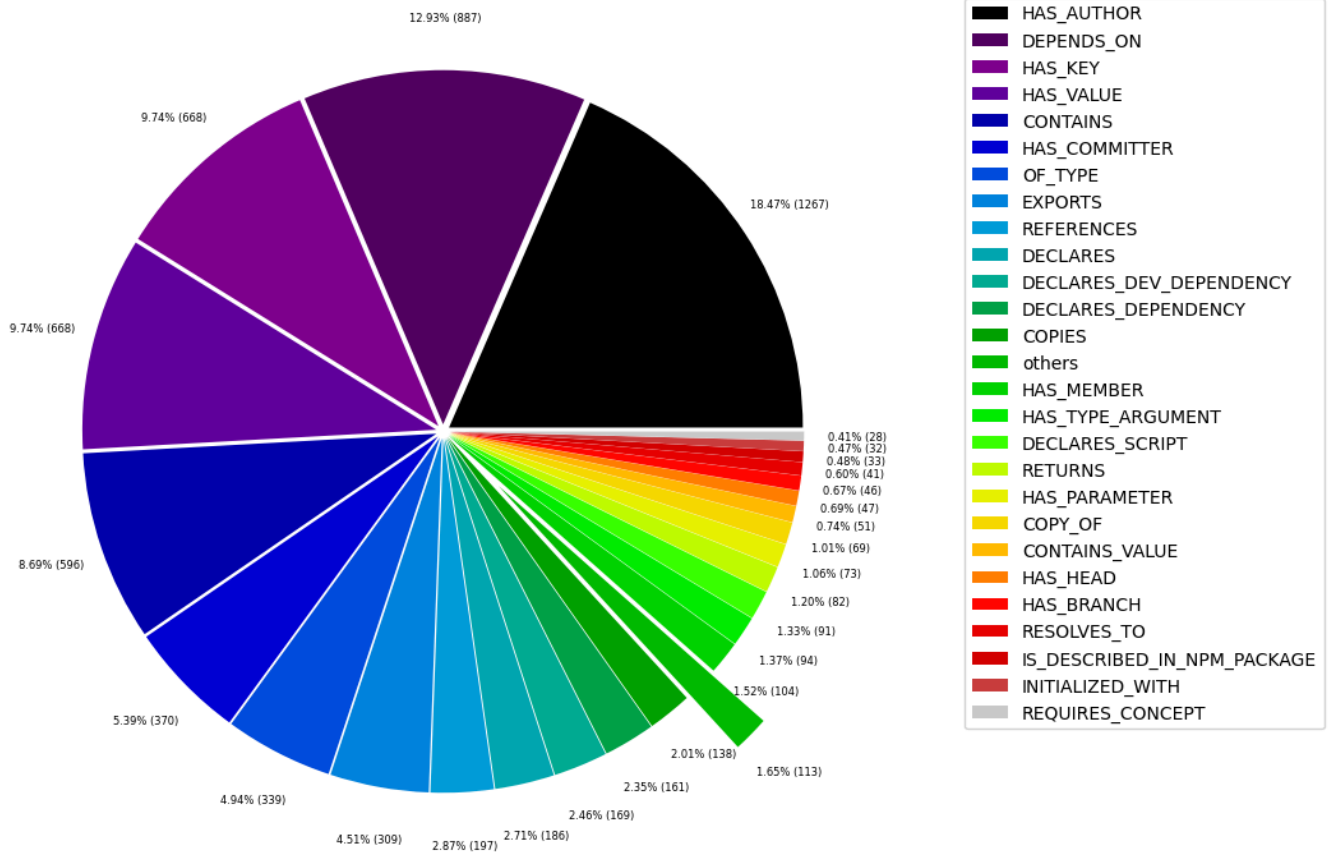
	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	DECLARES_PUBLISH_CONFIG	1	0.000298
1	CONSTRAINED_BY	4	0.001190
2	IN_REPOSITORY	5	0.001488
3	REFERENCED_PROJECTS	5	0.001488
4	PARENT	6	0.001785
5	MEMBER	6	0.001785
6	HAS_ROOT	6	0.001785
7	HAS_NPM_PACKAGE	6	0.001785
8	HAS_CONFIG	6	0.001785
9	HAS_ARGUMENT	6	0.001785
10	EXTENDS	6	0.001785
11	DECLARES_ENGINE	6	0.001785
12	CONTAINS_PROJECT	6	0.001785
13	CALLS	6	0.001785
14	DECLARES_PEER_DEPENDENCY	8	0.002380
15	USES	11	0.003273
16	INCLUDES_CONCEPT	19	0.005653
17	REQUIRES_CONCEPT	28	0.008330
18	INITIALIZED_WITH	32	0.009520
19	IS_DESCRIBED_IN_NPM_PACKAGE	33	0.009818
20	RESOLVES_TO	41	0.012198
21	HAS_BRANCH	46	0.013685
22	HAS_HEAD	47	0.013983
23	CONTAINS_VALUE	51	0.015173
24	COPY_OF	69	0.020528
25	HAS_PARAMETER	73	0.021718
26	RETURNS	82	0.024396
27	DECLARES_SCRIPT	91	0.027073
28	HAS_TYPE_ARGUMENT	94	0.027966
29	HAS_MEMBER	104	0.030941

Chart 2b - Lowest relationship count by type

Shows the lowest (less than 0.5% overall) relationship types. This plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Relationship types (less than 0.5% overall)



Node labels with their relationships

Table 3a - Highest relationship count by node labels and relationship type

Lists the 30 node labels and their relationship types with the highest number of occurrences.

	sourceLabels	relationType	targetLabels	numberOfRelationships	numberOfNodesWithSameLabelsAsSource	numberOf
0	[Git, Update, Change]	MODIFIES	[File, Git]	58793	58793	
1	[Git, Update, Change]	UPDATES	[File, Git]	58793	58793	
2	[Git, Commit]	CONTAINS_CHANGE	[Git, Update, Change]	58793	11298	
3	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Create]	17294	11298	
4	[Git, Change, Create]	MODIFIES	[File, Git]	17294	17294	
5	[Git, Change, Create]	CREATES	[File, Git]	17294	17294	
6	[Git, Commit]	HAS_PARENT	[Git, Commit]	12383	11298	
7	[Repository, File, Git]	HAS_COMMIT	[Git, Commit]	11298	1	
8	[Author, Git, Person]	COMMITTED	[Git, Commit]	11298	1267	
9	[Committer, Git, Person]	COMMITTED	[Git, Commit]	11298	370	
10	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Delete]	8782	11298	
11	[Git, Change, Delete]	DELETES	[File, Git]	8782	8782	
12	[Git, Change, Delete]	MODIFIES	[File, Git]	8782	8782	
13	[Repository, File, Git]	HAS_FILE	[File, Git]	5905	1	
14	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Rename]	3355	11298	
15	[Git, Change, Rename]	DELETES	[File, Git]	3355	3355	
16	[Git, Change, Rename]	RENAMES	[File, Git]	3355	3355	
17	[Git, Change, Rename]	MODIFIES	[File, Git]	3355	3355	
18	[Git, Change, Rename]	CREATES	[File, Git]	3355	3355	
19	[File, Git]	HAS_NEW_NAME	[File, Git]	1771	5905	
20	[Repository, File, Git]	HAS_TAG	[Git, Tag]	1758	1	
21	[Git, Tag]	ON_COMMIT	[Git, Commit]	1758	1758	
22	[Repository, File, Git]	HAS_AUTHOR	[Author, Git, Person]	1267	1	
23	[Json, Value, Object]	HAS_KEY	[Json, Key]	668	133	
24	[Json, Key]	HAS_VALUE	[Json, Value, Scalar]	552	668	
25	[Repository, File, Git]	HAS_COMMITTER	[Committer, Git, Person]	370	1	
26	[TS, Function]	DEPENDS_ON	[TS, ExternalDeclaration]	293	47	
27	[File, TS, Local, Module]	DEPENDS_ON	[TS, ExternalDeclaration]	236	6	
28	[TS, ExternalModule]	EXPORTS	[TS, ExternalDeclaration]	215	11	
29	[Package, File, Json, NPM]	DECLARES_DEV_DEPENDENCY	[NPM, Dependency]	169	29	

Graph Density

total_number_of_nodes (vertices): 112564

total_number_of_relationships (edges): 336124

-> total directed graph density: 2.652799007454451e-05

-> total directed graph density in percent: 0.002652799007454451